Universität Bonn

Notes for the lecture

Functional Analysis and PDE

held by

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TEXed by

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Corrections and improvements

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Contents

Lecture **1**

0 Overview/Teaser **2**

1 Structures **3**
1.1 Topological spaces 3

Appendix **A**

List of definitions **B**

List of statements **C**

Index **D**

Lecture

Chapter 0

Overview/Teaser

[09.10.2024, Lecture 1]

1. What is FA?
   - Analysis on infinite-dimensional spaces

     examples: $C^0(X), C^\infty(U \subseteq \mathbb{R}^n), \ell_p = \{a : \mathbb{N} \to \mathbb{R} \mid \sum_{n=0}^\infty |a_n|^p < \infty\}$ $(1 \leq p \leq \infty)$
   - Combination of geometric, analytic and (linear) algebraic ideas
2. What is FA good for?
   - Geometry/Topology with manifolds, bundles, loop space
   - PDEs, scalar products
   - Algorithms/Numerics (approxmiation theory; errors)
   - infinite-dimensional representations
   - Mathematical Physics / Quantum Mechanics; spectral theory
   - Probability Theory (infinite products)
3. Why combine PA & PDE?
   - $u : U \subseteq \mathbb{R}^n \to \mathbb{R}$ as a C^2 function:

     $$\Delta u(x) = \sum_{i=1}^n \frac{\partial^2 u}{\partial x_i^2}. \tag{0.1}$$

     with $-\Delta u = f$.

     Classical approach: Given a potential $f \in C^2$, does there exist a solution and is that solution unique?

     Modern approach: $\Delta : X \to Y$ linear map between function spaces. e.g. $X = C^2(U), Y = C^0(U)$. Existance of solutions for all $f \in C^0$ is equiv to surjectivity of $f\Delta$ and uniqueness is equivalent to injectivity.

     Fun fact: C^2 is a bad space for Δ. But Δ is a good operator on many function spaces, e.g. $C^{2,\alpha}$ with $0 < \alpha < 1$, so D^2u is α-Hölder continuous, or for $W^{2,p}$ as Sobolev spaces (where $D^2u \in L^p$ for $1 < p < \infty$), where D^2 is the weak derivative
4. „Big things“
   - Spaces (of founctions) and convergence (norm, scalarproduct)
   - (Bounded) Linear operators, Hahn-Banach theorem, uniform boundedness, ...
   - Hilbert spaces, projections
   - Spectral theory (LA: $A : \mathbb{R}^n \to \mathbb{R}^n$ symmetric, so $\langle Ax, y\rangle = \langle x, Ay\rangle \Rightarrow \exists$ orthogonal basis of eigenvectors; we want something similar in infinite dimensions which makes diagonazing easier, and we can easily compute the inverse or powers of the operator)

Chapter 1

Structures

Generally, all vector spaces will be with $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$.

- on „infinite-dimensional“ spaces
- Convergence and neighborhoods, topological space
- Metric space (metric, triangle inequality, gives a *distance*)
- Normed space (vector space, $\|\lambda x\| = |\lambda| \cdot \|x\|$ for $\lambda \in \mathbb{R}$, gives a *length*)
- Scalar product space ($\|x\| = \sqrt{\langle x, x \rangle}$, gives a *length and an angle*)

1.1 Topological spaces

Let X a set and 2^X the power set.

Definition 1.1: Topological space, Topology

A *topological space* $(X, \mathcal{T})$ is a pair of a set X and subset $\mathcal{T}$ of 2^X with the following properties:

(i) $\emptyset \in \mathcal{T}, X \in \mathcal{T}$

(ii) finite intersections: $U, V \in \mathcal{T} \Rightarrow U \cap V \in \mathcal{T}$

(iii) arbitrary unions: $U_\alpha \in \mathcal{T}\ \forall \alpha \in A \Rightarrow \bigcup_{\alpha \in A} U_\alpha \in \mathcal{T}$.

The set $\mathcal{T}$ is called a *Topology on* X and the elements in $\mathcal{T}$ are often called *open sets*. A set C is called *closed* if $X \setminus C$ is open.
If $\mathcal{T}_1$ and $\mathcal{T}$ are topologies on X, then $\mathcal{T}_1$ is called *finer* than $\mathcal{T}_2$ or $\mathcal{T}_2$ is called *coarser* than $\mathcal{T}_1$ if $\mathcal{T}_1 \supset \mathcal{T}_2$.

Example 1.2. (i) The finest topology is $\mathcal{T} = 2^X$.

(ii) The coarsest topology is $\mathcal{T} = \{\emptyset, X\}$.

(iii) Standard topology on $\mathbb{R}$: $U \in \mathcal{T}_{st} \iff \forall x \in U \exists \varepsilon > 0 : (x - \varepsilon, x + \varepsilon) \subseteq U$. We will usually use this topology.

(iv) Let $X = \mathbb{R}$ and $\mathcal{T} = \{U \subseteq \mathbb{R} \mid U = \emptyset \text{ or } U \text{ is countable}\}$. Then, the closed sets are countable[1] or $\mathbb{R}$.

(v) Let $(X, \mathcal{T})$ be a topological space. Then, for $A \subseteq X$, the *relative topology* $\mathcal{T}_A$ on A is defined by $\mathcal{T}_A = \{U \cap A \mid U \in \mathcal{T}\}$.

(vi) Let B be a set and $\mathcal{T}_\beta$ a topology on X for all $\beta \in B$. Then, $\mathcal{T} := \bigcap_{\beta \in B} \beta$ defines a topology on X.

[1] We call a set *countable* if it is empty, finite or has an bijection to $\mathbb{N}$.

Given any $\mathcal{S} \subseteq 2^X$,

$$\bigcap_{\substack{\mathcal{T} \text{ top.} \\ \mathcal{T} \supseteq \mathcal{S}}} \mathcal{T} \tag{1.1}$$

is a topology and is called the *coarsest topology containing* $\mathcal{S}$.

(*vii*) *Product topology* of $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ as topological spaces is definied with

$$\mathcal{S} := \{U \times Y \mid U \in \mathcal{T}_X\} \cup \{X \times V \mid V \in \mathcal{T}_Y\}. \tag{1.2}$$

The product topology is $\mathcal{T}_{\mathcal{S}}$, the coarsest topology containing $\mathcal{S}$.

The product topology is the coarsest topology so that the projections $\pi_1(x, y) = x$ and $\pi_2(x, y) = y$ are „continuous“.

Definition 1.3: Interior, Closure, Boundary

Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$. We define the *interior* A°, the *closure* $\overline{A}$ and the *boundary* ∂A by

$$A^\circ = \bigcup_{\substack{U \subseteq A \\ U \in \mathcal{T}}} U, \quad \overline{A} = \bigcap_{\substack{C \supseteq A \\ X \setminus C \in \mathcal{T}}} C, \quad \partial A = \overline{A} \setminus A^\circ. \tag{1.3}$$

Definition 1.4: Dense subsets

$A \subseteq X$ is *dense* if $\overline{A} = X$.

Definition 1.5: Open neighborhood

Let $x \in X$. Then, $U \subseteq X$ is an *open neighborhood of* x if $U \in \mathcal{T}$ and $x \in U$.

Definition 1.6: Continuous maps, Homeomorphisms

Let $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ topological spaces. Then, $f : X \to Y$ is called *continuous* if

$$\forall V \supseteq \{x\} \in \mathcal{T}_Y : \quad f^{-1}(V) = \{x \in X \mid f(x) \in V\} \in \mathcal{T}_X. \tag{1.4}$$

We call f a *homeomorphism* if f is bijective and both f and f^{-1} are continuous.

If $\mathcal{T} = 2^X$, and $f : X \to Y$ is continuous.

If $\mathcal{T} = \{\emptyset, \mathcal{T}\}$, then every constant map is continuous.

[09.10.2024, Lecture 1]
[11.10.2024, Lecture 2]

Appendix

List of definitions

1.1 Topological space, Topology 3
1.3 Interior, Closure, Boundary 4
1.4 Dense subsets 4
1.5 Open neighborhood 4
1.6 Continuous maps, Homeomorphisms 4

List of statements

Index

closed set, 3
continuous, 4
countable, 3

homeomorphism, 4

open neighborhood, 4
open set, 3

set
    boundary, 4
    closed -, 3
    closure, 4
    countable, 3
    dense, 4
    interior, 4
    open -, 3
    open neighborhood, 4
Space
    Topological -, 3

Topological Space, 3
Topology, 3
topology
    relative -, 3